

Review

Load-induced human skeletal muscle hypertrophy: Mechanisms, myths, and misconceptions

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Abstract

Mechanical tension is widely recognized as the primary stimulus underlying the molecular mechanisms that influence muscle hypertrophy induced by resistance training. Despite this, several outdated or overstated concepts continue to persist, both in the scientific literature and in the practical application of resistance training coaching and program design. Claims that acute hormonal responses, metabolic stress, cell swelling or “the pump” meaningfully contribute to hypertrophy are not supported by scientific evidence. Additionally, the concept of sarcoplasmic hypertrophy as a distinct and functionally meaningful contributor to hypertrophy lacks strong evidence. In this review, we critically evaluate several persistent misconceptions and contrast them with evidence-based mechanistic insights into load-induced hypertrophy. Specifically, we discuss the role (or lack thereof) of systemic hormones, metabolites, and cell swelling in promoting muscle hypertrophy. We also critically review the concept of sarcoplasmic hypertrophy and propose that it is not a meaningful contributor to muscle hypertrophy. Lastly, to translate knowledge for trainees and coaches, we discuss the upper limit of muscle hypertrophy and provide readers with evidence-based, reasonable expectations for muscle hypertrophy. We aimed, through this review, to use scientific evidence to enhance our understanding of what drives muscle hypertrophy and provide an evidence-based framework for resistance exercise training.

Keywords: Mechanical tension; Muscle mass; Resistance training; Skeletal muscle

1. Introduction

Resistance-exercise training (RET)-induced skeletal muscle hypertrophy results from a complex and incompletely understood interplay between external (e.g., RET, RET variable programming, diet, and sleep) and internal biological variables (e.g., genomic mechanisms, mechanotransduction, ribosomes, gene expression, epigenetic modification, satellite cell activity).^{1,2} For more (and deeper) information than we provide here, we direct the reader to recent comprehensive reviews on these topics.^{1,2} Skeletal muscle hypertrophy, defined as an increase in the axial cross-sectional area (CSA) of a muscle or muscle fiber, may be assessed using multiple methodologies, including magnetic resonance imaging (MRI), computed tomography, ultrasound, and/or biopsies examining muscle fiber CSA (fCSA).³ Of the potential “drivers” of

RET-induced hypertrophy, mechanical tension through progressive overload with RET is the most potent non-pharmacological stimulus and external variable for increasing skeletal muscle mass.⁴ This process requires stimulating muscle protein synthesis (MPS) to chronically exceed the rate of muscle protein breakdown (MPB) within a muscle fiber, thus leading to a net positive protein balance that ultimately promotes muscle protein accretion and eventual hypertrophy.⁵ The determinants of MPS are multifaceted, but largely involve a common nexus, which is the mechanistic target of rapamycin complex 1 (mTORC1)-dependent and mTORC1-independent mechanisms (e.g., the mitogen-activated protein kinase (MAPK)/extracellular signal-regulated kinase 1/2 (ERK1/2) pathway). These pathways are critical for increasing the rate of initiation of protein translation and/or elongation, ultimately leading to the accrual of proteins, a positive net protein balance, and muscle hypertrophy.² The most robust evidence-based mechanistic regulator of MPS is a mechanotransductive process, which is the result of mechanical tension (e.g., RET)

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that is subsequently “sensed” by mechanosensors such as $\alpha_7\beta_1$ -integrin and focal adhesion kinase,^{6,7} triggering a cascade of intracellular events that when summed over time lead to muscle hypertrophy; however, the precise mechanisms remain to be elucidated.^{1,8}

Additionally, although beyond the scope of this review, ribosomal biogenesis enhances the muscle’s ability to synthesize proteins, which plays a key role in translational capacity and thus hypertrophy.⁹ However, the need for ribosomal biogenesis to support RET-induced hypertrophy is unclear. For example, evidence suggests that higher training volumes and early increases in ribosome-related gene expression are associated with greater hypertrophy, which may support an enhanced translational capacity to facilitate muscle hypertrophy.^{10,11} In contrast, recent work by Brown et al.¹² showed that the greatest increases in muscle mass were associated with lower changes in ribosome-related gene expression, suggesting that translational efficiency, rather than capacity (i.e., ribosome content), may be more important for long-term RET-induced hypertrophy. Similarly, Godwin and colleagues¹³ found no association between rDNA copy number and RET-induced hypertrophy, further suggesting that relative rDNA copy number may not be associated with anabolism and muscle hypertrophy. Moreover, satellite cells are muscle stem cells that fuse with muscle fibers, contributing to growth and repair^{2,14} that may contribute to load-induced hypertrophy;^{15,16} however, we do not provide an expansive discussion on the role of satellite cells in RET-induced skeletal muscle hypertrophy. Importantly, genetic and epigenetic modifications will also influence an individual’s muscle growth potential.²

A fundamental understanding of some of the factors outlined here can help optimize muscle-building strategies. We refer the reader to these recent reviews for a more comprehensive overview of these hypertrophic mechanisms.^{1,2} While the foundational mechanisms of RET-induced hypertrophy remain to be fully understood, several longstanding claims for mechanisms underpinning hypertrophy continue to persist in the face of tenuous supporting evidence and/or direct evidence that refutes them. Perpetuating mechanisms or paradigms that ostensibly mediate hypertrophy, but are non-evidence-based, does not advance understanding and often leads to confusion in resistance training programming, as it attempts to maximize outcomes. Specifically, the acute elevations in anabolic hormones, metabolic stress, and cell swelling (“the pump”) have been proposed as drivers of hypertrophy,^{17–19} with a significant influence on RET program design, despite a lack of strong supporting evidence. Similarly, the concept of sarcoplasmic hypertrophy as an RET-induced adaptation¹⁷ and, importantly, a contributor to muscle hypertrophy remains unsupported by robust evidence.

In this review, we critically examine these misconceptions and contrast them with well-established mechanistic insights into load-induced hypertrophy. We begin by addressing the role—or lack thereof—of acute post-exercise changes in systemic hormone concentrations in driving hypertrophy, incorporating sex-based differences into an understanding of exercise-induced anabolic responses. Next, we evaluate the purported contributions of intracellular metabolic stress and

cell swelling to hypertrophy and whether these adaptations are additive or redundant in promoting hypertrophy. We follow this with a critical review of sarcoplasmic hypertrophy and its purported role as an RET-induced adaptation to hypertrophy. Finally, we discuss the upper limits of muscle hypertrophy and provide realistic expectations for growth based on existing evidence. By clarifying these concepts, our goal was to help scientists, trainees, and practitioners advance their knowledge by using an evidence-based approach to question “popular” but unfounded notions that persist despite contrary evidence.

2. Mechanical tension

Mechanical tension refers to the force generated by muscle fibers during active contraction or passive stretch. This force is somehow “sensed” by muscle fibers and results in a cascade of signaling events. Specific tension can be defined as a force normalized to the area of the muscle (or muscle fiber) in which it acts, with units expressed as Newtons per square meter or pascals.¹⁸ Mechanical tension is widely regarded as the most significant external factor driving the processes that underpin muscle hypertrophy in response to mechanical overload. At a minimum, it is responsible for initiating the intracellular signaling cascades related to hypertrophy following RET.¹⁹ It has been shown to independently stimulate mTOR,²⁰ possibly through the activation of the extracellular signal-regulated kinase/tuberosus sclerosis complex 2 (ERK/TSC2) pathway.²¹ In addition, these actions may be mediated via the synthesis of phosphatidic acid by phospholipase D.^{20,22} Phosphatidic acid may trigger the phosphorylation of p70S6K independently of mTORC1,²³ which provides an additional mechanism for promoting MPS. However, this mechanism has not been confirmed in humans, and more robust evidence is needed in this area. For a more comprehensive review and detailed schematic of the mechanically-induced activation of mTORC1, we refer the reader to this recent review.²⁴

Mechanosensors appear to be sensitive to the stimulus imposed on the muscle. Stretch-induced mechanical loading appears to elicit longitudinal accumulation of sarcomeres (i.e., in series), whereas more dynamic muscular actions (i.e., RET) increase CSA in parallel.²⁵ These adaptations suggest that the hypertrophic response may vary depending on the type of muscle action performed,²⁵ highlighting the complexities of the mechanosensors and their capacity to distinguish between different types of mechanical stimulus and promote corresponding adaptations.

Importantly, from an applied standpoint, mechanical tension and thus skeletal muscle hypertrophy can be obtained across a wide range of loads,²⁶ allowing for versatility in program design.

3. Mechanotransduction

Mechanical tension via RET provides the external stimulus required to activate mechanosensitive pathways that translate skeletal muscle loading, initiating adaptive responses that lead to muscle protein accretion. Force transmission within muscle occurs both longitudinally along the fiber length and laterally

through the surrounding fascia matrix.²⁷ The process of mechanotransduction describes how mechanical forces within muscle are converted into molecular signals that regulate intracellular anabolic and catabolic pathways.^{1,28} While the precise mechanisms of how skeletal muscle senses mechanical stimuli and initiates hypertrophy remain to be fully elucidated,^{1,8} several protein complexes have been identified as key mechanosensors. These molecular transducers play a crucial role during muscle contraction, converting mechanical signals into biochemical pathways that regulate muscle mass and function. The extracellular matrix (ECM) is thought to be central to this process, facilitating signal transduction that ultimately governs skeletal muscle hypertrophy.^{1,8}

It has been hypothesized that the $\alpha_7\beta_1$ -integrin plays a role in load-induced muscle hypertrophy via mTORC1-independent mechanisms.⁶ Among the primary mechanosensors, integrins play a critical role in intracellular signaling as part of focal adhesion complexes, also known as costameres. These sarcolemma-associated protein structures anchor the ECM and the sarcolemma at the Z-disk.¹ Focal adhesion complexes can directly enhance protein translation via ribosomal proteins, and their disruption impairs intracellular anabolic signalling.²⁹ There is some evidence to suggest that focal adhesion kinase (FAK), an enzyme localized within costameres, plays a key role in initiating these signals.²⁹ FAK expression is upregulated during mechanical overload, reinforcing its function as a mechanoreceptor in RET-induced hypertrophy.³⁰ However, Crossland et al.²⁹ showed using insulin-like growth factor-1 (IGF-1) *in vitro* that FAK inhibited IGF-1-mediated MPS and myotube hypertrophy, which limits the translation of these findings *in vivo*. Additionally, while mechanical stimulation may increase FAK activity in a variety of cells, there is currently no evidence to demonstrate a direct link between $\alpha_7\beta_1$ -integrin and FAK in skeletal muscle, nor activation of FAK by the integrin, and this area remains largely unexplored in human skeletal muscle.^{6,24} Thus, while FAK appears responsive to mechanical stimulation, mechanotransduction mechanisms remain to be fully understood and warrant further investigation.

In addition to integrins and focal adhesion complexes, other proteins have been implicated in hypertrophic adaptations. Notably, titin, a large elastic protein, has been identified as a primary contributor to passive force generation during eccentric contractions.³¹ Titin contains a stretch-activated kinase domain that responds to sarcomere elongation by exposing amino acid residues in its adenosine triphosphate (ATP)-binding pocket, thereby activating its kinase function.³²

Costameres play a critical role in force transmission from the sarcomere to the ECM.⁸ In response to mechanical stimuli, phospholipase C γ 1 colocalizes with FAK, catalyzing the conversion of phosphatidylinositol 4,5-bisphosphate into phosphatidic acid (PA) in human embryonic kidney 293 cells (HEK293T).^{33,34} In turn, PA activates the Hippo pathway effectors, including Yes-associated protein 1 (YAP) and transcriptional coactivator with PDZ-binding motif (TAZ), through a signaling cascade. These effectors are well-established regulators of cell growth in *Drosophila melanogaster* and various

mammalian tissues^{35,36} and have been shown to modulate myoblast proliferation and differentiation.^{37,38} While the direct link between YAP and mTORC1 remains unclear, animal studies indicate that YAP activation can sustain muscle growth even in the presence of mTORC1 inhibition via rapamycin.^{37,39} Additionally, YAP and TAZ appear to contribute to mechanically induced anabolic signaling by upregulating the gene expression of Slc7a5 and Slc3a2, which encode leucine transport proteins. These results suggest that mechanically loaded muscle may exhibit heightened sensitivity to leucine-stimulated muscle protein synthesis.⁴⁰

Furthermore, PA may influence muscle hypertrophy by modulating the Hippo signaling pathway and activating mTORC1.²⁰ Evidence from rodent muscle incubation and *ex vivo* stretch models suggests that mechanical stimuli can increase localized PA accumulation and suppress PA production, which ablates markers of mTORC1 activity following mechanical overload.²⁰ Collectively, these findings underscore the necessity of costamere-based protein sensors in hypertrophic signaling during the acute post-exercise period.

Despite significant progress, our understanding of mechanotransduction in RET-induced hypertrophy remains incomplete. Nevertheless, mechanical tension is widely recognized as the primary driver of muscle growth,² but the undiscovered interplay between different molecular sensors and signaling pathways presents exciting opportunities for future research. Further investigations are needed to elucidate how these factors collectively regulate mechanotransduction and skeletal muscle hypertrophy.

4. Hormones

It has been posited that both acute (i.e., minutes, hours) and chronic (i.e., days, weeks, months) fluctuations in systemic anabolic hormones, such as testosterone (T), growth hormone (GH), and insulin-like growth factor-1 (IGF-1), are internal variables with a causative role in RET-induced muscle hypertrophy.^{41,42} On the contrary, previous research that has directly investigated this thesis has yielded no support for the concept that acute RET-induced changes in these anabolic hormones mechanistically underpin increases in MPS or skeletal muscle hypertrophy.

Wilkinson and colleagues⁴³ employed a unilateral RET model in 10 young males who trained one leg thrice weekly for 8 weeks. Despite no changes in systemic hormone concentrations (free and total testosterone, GH, IGF-1, cortisol) following acute exercise, significant hypertrophy was observed only in the trained leg, with muscle CSA (by CT) and type IIx and IIa muscle fiber CSA increasing by 22% and 13%, respectively ($p < 0.05$). Strength also improved in the trained leg (leg press: +25 kg; knee extension: +21 kg, both $p < 0.05$), whereas the untrained leg remained unchanged, indicating that RET-induced hypertrophy can occur without any change in systemic hormonal changes. Alternatively phrased, acute post-exercise rises in T, GH, and IGF-1—ostensible “anabolic hormones”—are not needed for muscle hypertrophy to occur.

Subsequent work by West et al.⁴⁴ investigated whether acute RET-induced elevations in testosterone, GH, and IGF-1 enhanced myofibrillar protein synthesis (MPS) and intracellular anabolic signaling. Participants performed unilateral elbow flexor RET under conditions designed to elicit either a minimal or a substantial rise in anabolic hormones. Despite significantly (4–5 fold) higher anabolic hormone levels in the high-hormone (HH) condition, no differences in MPS rates or phosphorylation of key signaling proteins (e.g., p70S6K) were observed between HH and low-hormone (LH) conditions, leading the authors to conclude that acute hormonal elevations do not enhance fed-state anabolic signaling or MPS following RET.⁴⁴ The same acute design was extended to a 15-week unilateral training protocol to measure the effects of the elevated hormones chronically.⁴⁵ This study involved training one arm (elbow flexors) under high hormone (elevated T, GH, and IGF-1) conditions by following an intense leg-based resistance exercise protocol, where the contralateral limb performed only elbow flexor exercises and was thus persistently exercised under a low-intensity environment. Despite consistently greater post-exercise hormone elevations that the HH arm was exposed to, both conditions resulted in similar increases in muscle CSA (LH = 12% ± 2% vs. HH = 10% ± 2%, $p > 0.05$) and strength gains. These results refute the thesis that acute post-exercise systemic hormonal spikes influence hypertrophy or strength adaptations in young males.¹⁶

Corroboration for the lack of impact of the acute post-exercise rise in anabolic hormones comes from several larger studies.^{46,47} West and Phillips⁴⁶ examined the post-exercise elevations in GH, free testosterone, and IGF-1 in a relatively large cohort ($n = 56$). They reported that none of the rises in these hormones were significantly correlated with gains in lean body mass or improvements in leg press strength, confirming that transient hormonal fluctuations are not correlated with RET-induced hypertrophy or strength gains.

If systemic hormonal concentrations were primary regulators of hypertrophy, one would expect sex-based differences in RET-induced muscle growth, given that males have markedly higher circulating testosterone levels than females.⁴⁸ However, research continues to provide strong evidence that biological sex does not significantly impact relative RET-induced gains in lean mass. Despite having 10- to 20-fold lower total testosterone and up to 200-fold lower free testosterone concentrations than males following puberty,⁴⁸ multiple meta-analyses have reported similar relative increases in muscle mass and strength following RET.^{49,50} Additionally, mechanistic research from our lab showed no sex-based differences in muscle protein synthesis rates over 24 h following RET when males and females were compared in the fed state.⁵¹ While it has been speculated that estrogen may play a compensatory role in females, counterbalancing the lower testosterone levels,^{52,53} direct evidence for such a mechanism currently does not exist in the literature. It is also notable that postmenopausal females can gain, relatively, similar amounts of muscle to age-matched men.⁵⁴ Thus, it is clear that the acute hormonal rises do not affect protein turnover, signaling, or hypertrophy in males or females. Importantly, the longer-term exposure to

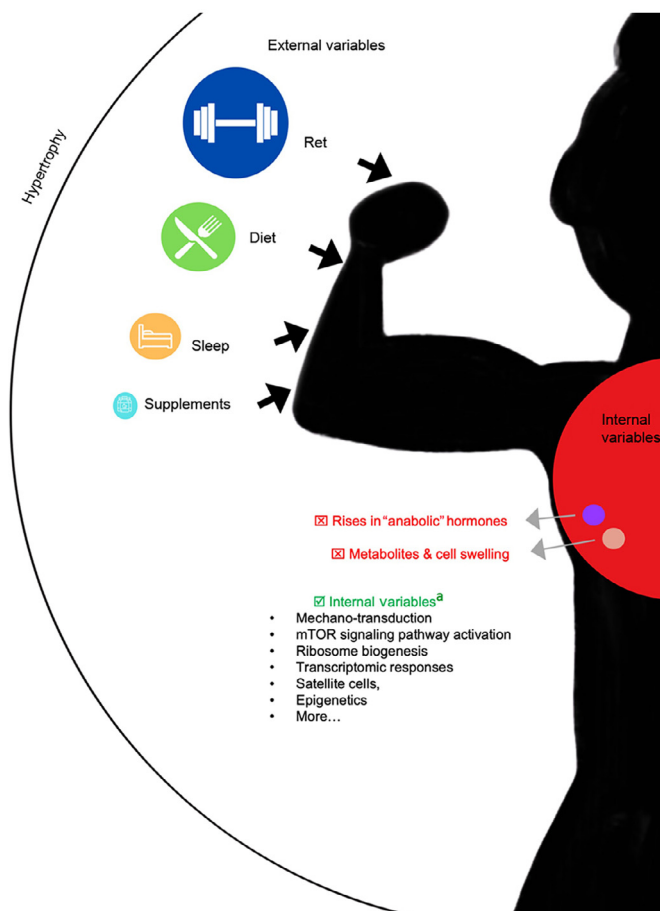


Fig. 1. Conceptual model of external and internal factors influencing skeletal muscle hypertrophy. Resistance exercise training-induced muscle hypertrophy is mediated by the interplay of both external inputs and internal biological processes. While external inputs are required to initiate adaptation, it is the initiation of internal mechanisms such as mechanotransduction, mTOR signaling, ribosome biogenesis, satellite cell activation, and epigenetic regulation that primarily determine the hypertrophic response. Acute physiological changes, such as transient hormonal elevations or metabolite-induced cell swelling, occur internally but are excluded from the model due to a lack of evidence supporting their causal role in muscle hypertrophy. ^a Internal variables are positioned outside the body silhouette for visual clarity; it denotes their connection to the internal hypertrophic processes. mTOR = mammalian/mechanistic target of rapamycin.

chronically lower testosterone and estrogen levels does not impair responsiveness to RET, which refutes the concept that systemic hormonal fluctuations within physiological ranges meaningfully influence RET-induced hypertrophy. It is important that we draw a distinction between natural endogenous hormones and exogenous administered hormones, particularly anabolic steroid use which is discussed further below. Fig. 1 illustrates a conceptual model of RET-induced hypertrophy, whereby external inputs (i.e., resistance training) act through internal processes such as mechanotransduction, mTOR signaling, ribosome biogenesis, satellite cell activity, and epigenetic regulation. However, changes such as systemic hormonal fluctuations and cellular swelling are absent due to their lack of evidence.

5. Metabolites

With the compelling evidence suggesting that muscle growth is primarily due to mechanical tension, another factor proposed to contribute to muscle hypertrophy from RET is the accumulation of metabolites, particularly lactate (La^-), inorganic phosphate (P_i), and hydrogen ions (H^+); collectively termed the metabolic stress theory of hypertrophy.^{55,56} Metabolic stress is maximized during exercise modalities that rely heavily, at least initially, on non-oxidative glycolysis for energy production, and RET would be a prime example, necessitating a reduction in phosphocreatine (PCr) levels to buffer against changes in cellular energy status. The imbalance between non-oxidative and oxidative glycolytic flux, and the entrance of pyruvate, via pyruvate dehydrogenase, into the tricarboxylic acid cycle and oxidative metabolism, results in an elevation in La^- , leading to lower pH levels.⁸ It is, however, notable that repeated all-out (high-intensity) sprints result in a sharp decline in reliance on non-oxidative (“anaerobic”) glycolysis, to the extent that oxidative disposal of pyruvate and oxidative generation of ATP become the predominant energy systems in muscle after as little as 2 all-out efforts. However, the origins of the concept that metabolites stimulate anabolic processes are difficult to locate. Nonetheless, in the context of RET, it has been proposed that intramuscular metabolite accumulation during lower-load RET may offset the lower mechanical tension by activating pathways that promote hypertrophy, resulting in a response comparable to that of high-load RET.^{57,58}

Research on bodybuilders has found that performing one set of 12 repetitions (reps) of biceps curls elevated intramuscular La^- to 91 mM/kg (dry weight) with values increasing to 118 mM/kg after 3 sets (resting La^- would be ~5 mM/kg dry weight).⁵⁹ The authors concluded that fatigue could have been partially caused by a decrease in PCr and an increase in H^+ in the first set, and by a progressively increasing H^+ (decreasing pH) in subsequent sets.⁵⁹ It has been proposed that fatigue could directly stimulate muscle hypertrophy through the accumulation of metabolites (i.e., La^- , P_i , H^+).^{56,58} Typical bodybuilding routines have been structured to capitalize on the potential growth-promoting effects of metabolic stress at the expense of higher load.^{55,60} Although the majority of bodybuilding routines perform ~8–12 reps per set with relatively short rest intervals⁶¹ eliciting greater levels of metabolic stress compared to heavier RET strategies,^{62,63} such as those often employed by powerlifters (~1–5 reps), both routines promote similar gains in hypertrophy when volume-load (sets $\times$ reps $\times$ load) is equated.^{64,65}

While several *in vitro* studies demonstrate that La^- enhances anabolic signaling and myogenesis,^{66–69} human research has found no causative role of pooling metabolites post-exercise. Valério and colleagues⁷⁰ investigated the association of serum metabolites following 8 weeks of high- and low-load training and observed significant correlations between changes in muscle thickness of the vastus lateralis muscle in the higher load group and levels of the metabolites. The authors also reported significant correlations between

changes in vastus lateralis muscle thickness and levels of the metabolites.⁷⁰ While the correlations seemed to be related to characteristics of the activated muscle fibers, the metabolic demand of the training protocols used and the process of MPS, it is important to note that the correlations observed do not imply causation and caution is warranted when drawing inferences on whether these metabolites play a role in hypertrophic adaptations or are merely associated with growth. We note that it is likely impossible to decipher causality in this scenario, as all muscular contractions are associated with metabolite accumulation; however, we note that not all contractions result in hypertrophy. While some literature supports a direct anabolic role of metabolites, none of these studies were specifically designed to isolate this effect independently of muscle contraction.⁷¹

Preclinical models have been employed to elucidate the effects of contraction-related metabolites on stimulating hypertrophy. For example, a study in male mice, in which oral La^- (100 mg/kg) and caffeine (36 mg/kg) were administered during 4 weeks of treadmill running, reported greater increases in muscle weight of the gastrocnemius and tibialis anterior compared to exercise and sedentary controls.⁶⁷ The authors speculated that La^- may be anabolic via alterations in protein expression, including increases in myogenin (involved in the regulation of SC differentiation) and a decrease in myostatin (a negative regulator of muscle growth), as well as an increase in follistatin (a myostatin inhibitor). However, the inclusion of caffeine largely confounds these findings (via increased intracellular calcium, among other mechanisms), as these compounds were not analyzed independently, making it unclear whether La^- alone influenced hypertrophy.⁶⁷

It has also been recently hypothesized that skeletal muscle protein lactylation, whereby lactate is used as a substrate to post-translationally modify cytoplasmic and nuclear enzymes, significantly contributes to a variety of exercise adaptations including skeletal muscle hypertrophy.⁷² However, direct supporting evidence is also lacking. In this regard, recent research from Mattingly and colleagues⁷³ combined a human and rodent model, challenging the hypothesis that La^- makes a meaningful contribution to skeletal muscle hypertrophy through protein lactylation. In young men, an acute bout of 8 sets of an exhaustive lower-body session significantly elevated La^- levels ~7.2-fold ($p < 0.001$); however, this did not alter cytoplasmic or nuclear protein lactylation, nor did it affect the expression of lactylation-responsive genes. Similarly, 6 weeks of resistance training in humans increased the muscle size of the vastus lateralis (+3.5%, $p = 0.04$) without significant changes being reported in vastus lateralis (VL) cytoplasmic protein lactylation. Moreover, in a unilateral crossover design,⁷⁴ researchers infused either saline or sodium lactate during resistance exercise to examine a potential signaling role. Post-exercise blood lactate concentrations were ~130% higher in the lactate trial (3.0 mmol/L vs. 7.0 mmol/L, $p < 0.001$), whereas intramuscular levels increased only modestly (27 mmol/kg vs. 32 mmol/kg dry weight, $p = 0.003$). Despite these systemic differences, lactate infusion did not alter intramuscular pH, exercise-induced phosphorylation of

mTOR, S6K1, or, p44, nor did it affect fractional protein synthesis (FSR) rates during 24 h of recovery.⁷⁴ In mice, 10 days of mechanical overload led to substantial hypertrophy of the plantaris muscle (+43.7%, $p < 0.001$); however, there was no difference in cytoplasmic protein lactylation between groups ($p = 0.369$), and nuclear lactylation was reduced compared to Sham mice ($p < 0.001$). These null findings suggest that La^- accumulation does not appear to mediate hypertrophic adaptations via lactylation pathways, and they collectively question the purported anabolic role of La^- in skeletal muscle RET-induced hypertrophy.⁷³

It has been hypothesized that metabolic stress contributes to hypertrophy through indirect downstream mechanisms, with higher levels of metabolic stress resulting in greater cell swelling and systemic hormone release. The thesis is then that such stress and hormones stimulate satellite cell activation and proliferation, supporting anabolism while suppressing catabolic processes.^{56,75} In addition, since increases in La^- , hydrogen ions, and inorganic phosphate, as well as reductions in phosphocreatine, are all elevated following muscle contraction, it has been posited that a reduced blood pH may promote hypertrophy via growth hormone release and increased motor unit recruitment to maintain force production.⁷¹ However, given that the acute rise in systemic anabolic hormones (including growth hormone) following RET is not correlated with MPS or hypertrophy⁷⁶ this mechanism is unlikely. Rather, metabolic stress may influence muscle fiber recruitment as muscle fibers fatigue and metabolite accumulation occurs, allowing other muscle fibers to be recruited to perform the task (i.e., complete the repetition), ultimately leading to the use of higher-threshold motor units to maintain force production.⁷¹ Thus, as RET-induced hypertrophy is, as we show above, largely attributable to mechanotransduction, it is possible that metabolites do not possess specific anabolic properties but rather augment muscle fiber activation, leading to a mechanotransduction cascade of more muscle fibers, which may induce greater hypertrophy.⁷¹

Blood flow restriction (BFR) is an RET strategy used to create a relative hypoxic environment that increases metabolic stress and has been shown to elicit muscle hypertrophy, often using loads that fall between 20% and 30% one repetition maximum (1RM).⁷⁷ The localized metabolic stress caused by BFR has been hypothesized to be a mechanism as to how BFR induces hypertrophy with loads $< 30\%1\text{RM}$, which would otherwise be suboptimal under normal RET conditions.^{78–80} However, heavier load muscle contractions also result in relative hypoxia⁸¹ while arterial blood continues to try and deliver oxygen to the muscle, unlike what would occur with BFR. Thus, if metabolic stress were as directly additive to hypertrophy, adding BFR to regular RET should stimulate more growth; however, research does not support this proposition. Notably, a recent study by Bergamasco and colleagues⁸² employed BFR in conjunction with higher-load RET (80%1RM) and compared it to a volume-equivalent RET using 80%1RM only. Using a within-subject design, the BFR condition resulted in greater markers of metabolic stress; however, there were no differences in muscle hypertrophy,

suggesting that all else being equal, metabolic stress does not have an additive effect on muscle hypertrophy.⁸² Alternatively, BFR through impaired oxygen delivery would lead to premature metabolite-related fatigue, facilitating the earlier recruitment of higher-threshold motor units and the activation of otherwise inactive fibers, which would subsequently lead to hypertrophy. Such motor unit recruitment would be unlikely to occur without BFR unless numerous repetitions were performed. Even so, the balance of evidence does not show any convincing data of a direct effect of metabolic stress on signaling protein activation, stimulation of anabolic processes, and hypertrophy in humans. Rather, there is evidence that BFR results in premature muscular fatigue that leads to earlier recruitment of higher threshold (type II fiber innervating) motor units (MU),⁸³ suggesting that adaptations are likely due to the mechanical loading of these fibers and not of any metabolite accumulation per se. Additionally, MPS is not stimulated when BFR is applied independent of RET,⁸⁴ nor is BFR sufficient to prevent atrophy during periods of bed rest.⁸⁵ If a BFR-mediated metabolite-mediated stimulation influence on metabolism does exist, the additive benefit of metabolite accumulation in stimulating MPS is of minimal influence.

Overall, we contend that there is little to no evidence to suggest that any single or combination of metabolites meaningfully influences RET hypertrophy. It is important to acknowledge that there are over scores of metabolites in muscle and blood, which may be directly/indirectly associated with muscle hypertrophy.¹ However, exercise modalities that produce similar or greater metabolite concentrations, such as endurance, high-intensity interval training (HIIT), or sprinting,^{86,87} result in significantly less muscle hypertrophy (if any), compared to RET, which further supports the notion that metabolites do not meaningfully contribute to muscle growth.

5.1. Cell swelling: “The pump” does not contribute to RET-induced hypertrophy

Another popular proposed mechanism linking metabolic stress and hypertrophy is intracellular hydration (i.e., cell/muscle swelling).⁸⁸ Bodybuilders frequently refer to this phenomenon as “the pump”,⁸⁹ and it is often pursued in RET sessions by structuring workouts to maximize intracellular fluid accumulation. The pump has its origins in post-exercise hyperemia and the opening of muscular vascular and capillary beds, which, with increased flow, lead to an increase in muscle size. While aesthetically pleasing, the mechanism by which the pump affects hypertrophy is still difficult to establish. Acutely, the pump is proposed to result in increased interstitial fluid pressure, leading to cellular swelling, which has been demonstrated to function as a physiological regulator of cell function by stimulating protein accretion via an increase in MPS and reductions in MPB.⁸⁹ However, much of this mechanistic work has been conducted in non-muscle cells (i.e., hepatocytes), which lack the ECM lattice that would normally constrain myofibers. While it has been suggested that basic research offers the rationale that RET-induced cell swelling may enhance hypertrophy,⁸⁹ there are limited

human RET studies directly investigating this. It has been hypothesized that cellular swelling may increase hypertrophy via satellite cell activity;⁹⁰ however, this has yet to be demonstrated in humans.

Although there is a lack of strong evidence, 2 studies in humans have explored the relationship between hyperemia and hypertrophy.^{88,91} Hirono and colleagues⁸⁸ had participants perform leg extensions over 6 weeks, reporting that greater acute muscle swelling (measured via ultrasound) after one session of RET led to greater hypertrophy following 6 weeks of continued RET.⁸⁸ Correlations of muscle swelling and hypertrophy at 0 min, 5 min, 10 min, and 15 min were 0.441, 0.550, 0.547, and 0.475, respectively, suggesting that, in the absence of measuring any other mechanisms and using univariate correlation, ~25% of the variance in hypertrophy could be explained by cellular swelling. Oda and colleagues⁹¹ aimed to see if the swelling of the peroneus brevis and peroneus longus was associated with hypertrophy following 8 weeks of training. Participants performed 2 sets of 100 reps, 3 times weekly, and researchers assessed the correlation in changes of muscle swelling and hypertrophy following the intervention, reporting a significant positive correlation ($r = 0.682$) between muscle swelling and hypertrophy. However, this study was presented as an abstract and has not, to our knowledge, been published in full form.⁹¹ Moreover, the protocol of 100 reps limits generalizability to traditional RET regimens.

A key limitation of these studies is that they did not compare 2 different protocols to determine whether exercises that elicit a greater pump lead to superior hypertrophy. Furthermore, ultrasound measurements of cell swelling have limited practical application, as “the pump” is a subjective experience and does not accurately indicate the degree of actual cell swelling. Moreover, there is no evidence to validate an individual’s perception of a pump and/or correlate this with cellular swelling.

Van Vessel et al.⁹² conducted both acute and chronic studies in resistance-training novices to test whether muscle fiber typology—and its associated post-exercise blood flow (“hyperemia”)—predicts hypertrophy. Participants classified as “fast-typology” (FT; higher type II fiber %) showed 37%–60% greater femoral artery blood-flow area under the curve (AUC) over 2 h after a single bout at 40%–80% 1RM compared to “slow-typology (ST)” peers, with the largest difference at 60%1RM. Although acute post-exercise hyperemia was markedly greater in FT than ST individuals, this did not translate into greater long-term hypertrophy when both groups trained to failure at the same relative load. Thus, while increased muscle perfusion post-exercise can support amino-acid delivery and MPS, within the context of sufficient mechanical stimulus (training to failure at 60%1RM), differences in post-exercise hyperemia alone do not determine the magnitude of hypertrophy.

Given the lack of evidence supporting a direct hypertrophic benefit of metabolites or cell swelling, we conclude that “the pump” does not influence muscle hypertrophy. However, from an applied context, if an individual enjoys the sensation and finds that it promotes adherence to an RET regimen, we

recommend including it. That said, RET programming should not be designed with the primary goal of achieving a pump, as evidence suggests it has no impact on hypertrophy.

5.2. Sarcoplasmic hypertrophy: A critical look at non-myofibrillar expansion

Sarcoplasmic hypertrophy refers to a disproportionate expansion of the sarcoplasm (i.e., the proportion of myofiber that is not occupied by myofibrils) relative to myofibrillar protein accretion.¹⁷ Research has reported that muscle hypertrophy differs between bodybuilders and powerlifters, noting that powerlifters are significantly stronger than bodybuilders despite the fact that bodybuilders have large muscle mass and muscle CSA on par (or greater) than powerlifters, presumably due to differences in training methodologies.⁹³ Conceivably, increases in sarcoplasmic hypertrophy in bodybuilders may be a training-specific adaptation arising from the use of lighter-load, higher repetition modalities and would go to a reason why bodybuilders are not as strong (i.e., they gain non-myofibrillar-related/force-generating proteins in their muscle fibers). Such training may promote greater sarcoplasmic hypertrophy than heavier-load, low-repetition training, suggesting that increases in sarcoplasmic hypertrophy may augment muscle size without concomitantly increasing muscle strength. Roberts and colleagues¹⁷ have previously proposed 3 scenarios where sarcoplasmic hypertrophy coincides with RET adaptations: (a) a transient symptom of training-induced edema, (b) a transient mechanism leading to muscle fiber growth, and/or (c) an outcome of a myofibrillar protein accretion threshold being reached in well-trained individuals.

MacDougall et al.⁹⁴ was the first to provide sound evidence that sarcoplasmic hypertrophy may occur in the absence of myofibrillar accretion. Using transmission electron microscopy (TEM), the authors reported that 6 months of upper-arm RET in untrained males increased type II fiber CSA but was accompanied by a 3% reduction in the two-dimensional space occupied by myofibrils ($p < 0.05$) alongside a ~15% increase in sarcoplasmic volume. Notably, data from the same study with experienced lifters displayed larger type II muscle fibers compared to the untrained participants *prior to* the intervention, yet showed ~30% lower space occupied by myofibrils and a two-fold greater value in sarcoplasmic volume.⁹⁴

Building on this work, Haun and colleagues⁹⁵ extensively aimed to investigate sarcoplasmic hypertrophy using modern histological, biochemical and proteomic approaches. Thirty-one young men with prior RET experience were recruited to participate in a 6-week training program, which included progressive volumes, starting at 10 sets in the first week and scaling up to 32 sets in the final week. Fifteen participants who had notable hypertrophy in fiber cross-sectional area (~23%, 320–1600 μm^2) were further examined using biochemical, histological, and proteomic analyses to investigate the nature of their hypertrophy. The results suggested that in these fibers that while glycogen concentration remained unchanged, mitochondrial volume decreased. In addition, actin and myosin concentrations both decreased, whereas sarcoplasmic proteins

tended to increase. Proteomic analysis revealed that the expression of proteins involved in non-oxidative metabolism also increased. Of the 15 individuals, 7 underwent another biopsy 8 days after the cessation of training to determine how deloading affected protein biomarkers, and interestingly, these effects seemed to persist. These findings were the first to suggest that short-term, high-volume RET may elicit sarcoplasmic hypertrophy as opposed to myofibrillar hypertrophy, without contingent alterations in glycogen concentration.⁹⁵ However, a more conservative interpretation raises some skepticism around these findings. First, the repeated measures analysis of variance (ANOVA) for fCSA data was not significant ($p = 0.152$), and the concentrations of myosin and actin were not significant, with p values of 0.052 and 0.055, respectively. Nonetheless, since the values were approaching significance, the researchers performed a forced *post hoc* analysis to indicate that concentrations for both proteins decreased by ~30%.⁹⁵ It could be argued that this is not appropriate, and researchers should have reported the effect sizes and left the omnibus statistics as they were.

Furthermore, of the 15 individuals included, one participant had extremely high concentrations of actin and myosin pre-training (>150 AU/mg) compared to the others, who were all <100 AU/mg. Moreover, this participant experienced a significant decline following the first 3 weeks of training. When excluding these data, the decreases in myosin/actin no longer become significant (the p -value increased from 0.035 to 0.053), suggesting that the results could be a false positive, which is more common in research with small sample sizes. In addition, this population consisted of 15 participants from a larger sample of 31 young men, which raises the question of whether this would also be observed in the group of lower responders. If accepting these findings⁹⁵ at face value, and sarcoplasmic hypertrophy occurs as an RET adaptation, questions remain as to the mechanism, purpose and the impact on the adaptive process.

Exercise training promotes an increase in intramuscular glycogen stores (and thus fluid),⁹⁶ with research from MacDougall et al.⁹⁷ reporting an increase in glycogen concentrations of 66% compared to those at rest following 5 months of RET. Furthermore, bodybuilders have been reported to possess double the glycogen content compared to sedentary individuals,⁹⁸ which enzymatic alterations and a greater storage capacity in larger muscles may mediate. As 3–4 g of water is bound to each gram of glycogen,^{96,99} these alterations may promote fluid retention. This finding was observed by MacDougall and colleagues,⁹⁷ who found that training the triceps at a high volume led to an increase in the relative space taken up by glycogen, suggesting that glycogen may be contributing to sarcoplasmic “growth”. Moreover, a study from Ribeiro and colleagues¹⁰⁰ reported significant increases in intracellular water at both the midpoint and end of a 16-week progressive RET regimen (men = 8.2%; women = 11.0%), providing evidence that RET promotes an increase in muscle fiber hydration status.¹⁰⁰ This intervention employed a traditional bodybuilding-type routine of 3 sets of 8–12 reps taken to failure, with 60–90 s of rest between sets. Whether these training-induced increases in intracellular

hydration are specific to this style of training or are inherent to all RET training modalities remains to be elucidated.

It was once believed that energy systems utilized during a moderate-to-high repetition bodybuilding routine (i.e., non-oxidative glycolysis) differ from those used during a low-repetition powerlifting routine (i.e., ATP, PCr), providing the rationale for training-specific adaptations in the sarcoplasmic fraction; however, research on this notion remains scant, and the notion is unproven. As subsequent sets progress into a training session, 100% of the ATP generated can be attributed to oxidative phosphorylation. Previous work from our lab¹⁰¹ found that training at 90%1RM promoted greater sarcoplasmic protein synthesis compared to 30%1RM at 4 h post-exercise; however, by 24 h post-exercise, the 30%1RM leg had greater levels than the 90%1RM condition. While the two are related, it is important to note that these findings are specific to myocellular protein fractions and do not necessarily reflect long-term changes in hydration status. Additionally, this study¹⁰¹ was an acute intervention, and it is unknown whether these findings would have persisted over a longer duration. It should be noted that the provision of carbohydrate can significantly increase the relative space taken up by glycogen,⁹⁶ whereas immediately following hard resistance training, the relative space taken up by glycogen is decreased,¹⁰² suggesting that if glycogen were driving sarcoplasmic growth, it would be a transient and variable response. Thus, given the current evidence, it remains unclear whether glycogen is a significant driver of sarcoplasmic hypertrophy; however, it seems that it can occur in the absence of alterations in glycogen concentration.

Despite the work reporting sarcoplasmic hypertrophy, there is evidence in humans refuting it. Notably, Trappe et al.¹⁰³ had 7 untrained older males perform 12 weeks of RET and examined specific tension changes. They found that fCSA of type I and IIa fibers increased by 20% and 13%, respectively, with similar pre- and post-exercise tension values, suggesting proportional accretion of myofibrillar protein and fiber growth.¹⁰³ A follow-up study in 7 older untrained females found no hypertrophy of type II fibers and no effect on specific tension values, suggesting that the growth of type I fibers was accompanied by proportional accretion of myofibrillar protein.¹⁰⁴ These findings are in line with research from Widrick et al.,¹⁰⁵ who reported 12 weeks of RET elicited increases in hypertrophy of the vastus lateralis without alterations in isolated fiber tension values. Moreover, prior work by Roberts involved untrained males who had undergone 12 weeks of RET and clustered participants into higher and lower responders based on a composite score of changes in mean fCSA, dual-energy X-ray absorptiometry (DXA) total lean body mass, and vastus lateralis thickness. Thirteen high responders experienced a 34% increase in mean fCSA and a 24% in vastus lateralis thickness. Despite increases in muscle size, there were no changes in myosin heavy chain or actin protein abundance. These findings were also observed in the 12 non-responders, suggesting that myofibrillar protein accretion in high responders was proportional to fiber and tissue growth.¹⁰⁶ A more recent investigation from Vann et al.¹⁰⁷ had 15 previously trained men perform 4 sessions of RET per

week for 10 weeks, targeting 12–16 weekly sets for the quadriceps. Notably, type II muscle fCSA showed a significant increase of 19%, with no changes in sarcoplasmic protein content or percent fluid in muscle tissue observed. However, myosin heavy chain protein abundance trended downward, and actin protein abundance decreased significantly with training.¹⁰⁷ Additionally, Rupé and colleagues¹⁰⁸ reported that 10 weeks of full body RET (2 × weekly) in untrained males promoted a proportional expansion of the myofibrillar compartment. These findings, obtained using phalloidin-actin staining corroborate prior TEM data showing that ~80% of intracellular space within muscle fibers is occupied by myofibrils.^{94,108,109}

Multiple tracer studies have also contested the significance of sarcoplasmic hypertrophy contributing to long-term muscle hypertrophy. Notably, Moore et al.¹¹⁰ found that an acute bout of RET elicited a robust increase in fasting myofibrillar protein synthesis rates and minimal stimulation of sarcoplasmic protein synthesis rates up to 5 h post-exercise. In addition, they reported that sarcoplasmic protein synthesis rates were more sensitive to post-exercise nutrient provision rather than to the contractile stimulus itself.¹¹⁰ Subsequent work from Wilkinson employing the deuterium oxide (D₂O) tracer method over 8 days reported that 4-day of RET significantly increased myofibrillar protein synthesis with no effect on sarcoplasmic protein synthesis,¹¹¹ which is further supported by Brook et al.¹¹² who found elevated rates of myofibrillar protein synthesis following 6 weeks of RET using the same method. However, it is notable that Vann et al.¹¹³ utilized a within-subject study design in previously trained men to report that 6-week integrated non-myofibrillar protein synthesis rates were greater in a higher volume-trained limb versus higher-load trained limb despite both limbs presenting similar myofibrillar protein synthesis rates. Taken together, these results suggest that myofibrillar proteins accumulate and contribute to the increase in muscle fiber size at the onset of RET, which may coincide with the studies conducted on untrained participants.

Lastly, more recent evidence from Jorgenson and colleagues¹¹⁴ strongly refutes sarcoplasmic hypertrophy as an RET-induced adaptation. The researchers developed and validated a novel imaging method for visualizing myofibrils using

a standard fluorescence microscope and employed both animal and human models of increased mechanical loading (i.e., RET) to discover that the radial growth of muscle fibers is largely mediated by myofibrillogenesis.¹¹⁴ While the study does not completely rule out sarcoplasmic hypertrophy, it provides compelling evidence that structural adaptations are associated with mechanotransduction rather than metabolic or volumetric changes, which is in agreement with a previous review by Jorgenson and colleagues,¹¹⁵ which indicated that radial growth of myofibers is the primary contributor to growth in response to mechanical overload.

It is possible that sarcoplasmic hypertrophy, if it occurs, is transient and is backfilled with myofibrillar proteins, as has been suggested.¹¹⁶ Overall, significant knowledge gaps remain in human research regarding the confirmation of sarcoplasmic hypertrophy and its role in facilitating RET-induced adaptations. Importantly, training status is likely a key factor as most of the work demonstrating proportional increases in myofibrillar and sarcoplasmic expansion was conducted in untrained individuals, whereas the research from previously trained individuals may suggest disproportionate growth. It is possible that as a person becomes more trained and approaches a hypertrophic ceiling, the significance of sarcoplasmic hypertrophy in promoting long-term muscle hypertrophy is increased, but this notion remains speculative at this point. It may be a short-term adaptation in extreme cases, such as anabolic steroid use, where hypertrophy is rapid and robust, or where trained individuals significantly increase their training volume, but that remains speculative.¹¹⁵ Nevertheless, if sarcoplasmic hypertrophy indeed facilitates hypertrophy, it plays a minor role compared to myofibrillar protein accretion. Fig. 2 illustrates how the CSA of a muscle fiber can either increase through myofibrillogenesis or sarcoplasmic hypertrophy. Notably, it is plausible that they may occur synergistically.

5.3. The upper limit of muscle growth: How much muscle can you gain?

As shown in Fig. 3, early RET adaptations are strength gains largely mediated by neural and skill acquisition, while muscle hypertrophy becomes more significant over time

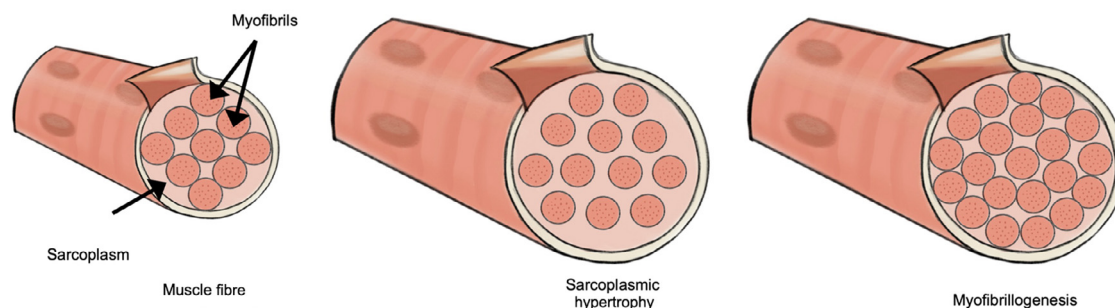


Fig. 2. Muscle fibre adaptations via myofibrillogenesis and sarcoplasmic hypertrophy. From left to right, we highlight the differences in how the cross-sectional area (CSA) of a muscle fibre may increase via resistance exercise training. Mechanical loading is hypothesized to stimulate the addition of myofibrils, leading to an increase in fibre CSA. While sarcoplasmic expansion (e.g., increases in glycogen, mitochondria, fluid, etc.) may contribute to fibre volume, the predominant and functionally meaningful adaptation is due to myofibrillogenesis.

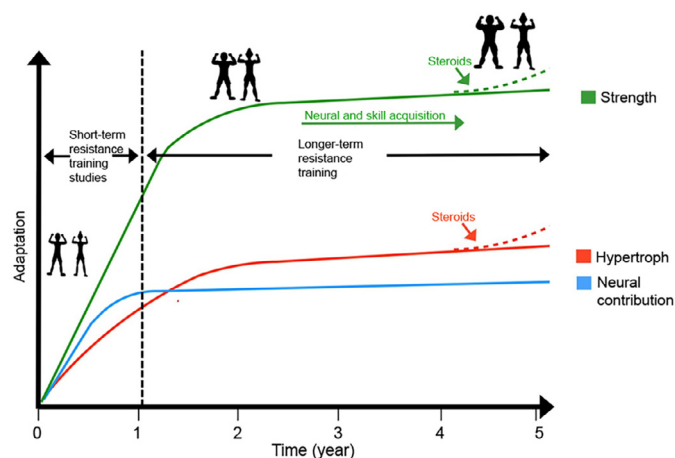


Fig. 3. Time course of neural and hypertrophic adaptations to resistance exercise training. This schematic highlights the relative contributions of neural, skill acquisition and muscle hypertrophy to resistance exercise training adaptations over a 5-year period. Neural factors primarily mediate early strength gains while hypertrophic adaptations play a progressively larger role over time. The figure also emphasizes that most resistance training studies are short-term and may not capture the full extent of long-term adaptations. This figure was adapted and created with inspiration from Ref.¹¹⁷

before eventually plateauing.¹¹⁷ Most training studies, however, are limited in that duration which typically range from 8 to 12 weeks, and therefore only capture only the initial onset of the adaptation curve presented. In practice, strength is a multifaceted skill and can be defined as the ability to produce force against an external resistance¹¹⁸ derived from a variety of factors included muscle hypertrophy, neural efficiency, and motor skill/technical proficiency and refinement (i.e., practice). Yet, we remain ignorant of the interplay and of how these variables contribute over the course of an individual's training career. In fact, while we contend that muscle fiber hypertrophy is contributory to strength changes, this thesis has been challenged.¹¹⁹ While evidence suggests that the most substantial muscle hypertrophy typically occurs within the first 3–6 months, after which the rate of increase declines substantially.¹²⁰ More long-term research is needed to better understand how hypertrophy, neural adaptations, and skill refinement interact to produce sustained strength gains over time.

Notably, Kraemer et al.¹²¹ observed an 11% increase in arm CSA among non-resistance-trained women during the initial 3 months of training. However, this rate of growth declined to 6% over the subsequent 3 months, despite continued training.¹²¹ Thus, as one becomes more experienced, the rate of muscle hypertrophy becomes increasingly difficult to measure and observe. For example, Alway et al.¹²² examined 10 highly competitive bodybuilders over 24 weeks and found no significant changes in muscle fiber size throughout the training period, highlighting the difficulty of achieving hypertrophy beyond the initial training phases, particularly in advanced trainees. Indeed, study duration is a significant limitation for research in trained individuals; nonetheless, there is good evidence for a dose–response relationship between

training volume and muscle hypertrophy.^{123,124} It is common to observe quadriceps thickness increases of 0.10–0.25 cm following a training intervention;¹²⁵ however, some research has reported changes as high as 0.72 cm over an 8-week intervention.¹²⁶ Thus, while a dose–response relationship appears to exist, it is highly variable and influenced by numerous factors, including genetics, previous training volume, rest intervals, the type of measurement used, and, to a limited degree, nutrition^{3,10,127,128} (see Fig. 1 for an overview of the interplay between the external and internal variables that induce skeletal muscle hypertrophy). It is essential to acknowledge that while research can help inform individuals about establishing appropriate training volumes, no single study should be used to draw strong inferences, as most studies are limited by their small sample sizes and typically involve untrained participants.

Longitudinal RET adaptations are scarce, given that the majority of studies range between 8 and 12 weeks, which is most likely dictated by the duration of an average college or university semester. Although participants in RET research are predominantly untrained or recreationally active, it is imperative to investigate the question: how much muscle can a person gain over their lifetime of training? Despite weak evidence to quantify inter-individual variation in RET responses for appropriate categorization,¹²⁹ Trexler¹³⁰ has estimated an average response alongside predictions for “higher” and “lower” responders as well as the response seen from anabolic steroid users. Forbes¹³¹ suggested that there is a theoretical natural upper limit to total fat-free mass (FFM) of 100 kg in males and 60 kg in females. However, as research in this area has expanded in recent years, particularly in athletic populations, these theoretical limits have been repeatedly refuted, with some studies reporting FFM totals exceeding 120 kg in males and 82 kg in females.^{132,133} More recently, a 2020 meta-analysis of 111 studies from Benito and colleagues¹³⁴ reported the effects of RET on a proxy of muscle hypertrophy (FFM, lean muscle mass, lean mass, and skeletal muscle mass) in male participants. FFM is of particular interest given its ease of measurement and utility for assessing height-normalized muscularity levels via the fat-free mass index (FFMI) calculated by dividing FFM (kg) by the square of an individual's height (in meters).¹³⁰

Additionally, FFM was the most commonly reported metric, encompassing a total of 61 studies across 82 groups, involving 951 participants (24 ± 3 years; 79.5 ± 6.4 kg; and 178 ± 4 cm; mean $\pm$ SD). RET interventions ranged from 4 weeks to 24 weeks in duration, with an average duration of 10.4 weeks (± 5.4 weeks). The average response to training was a gain of 1.53 kg (95% confidence interval (95%CI): 1.30–1.76 kg). Thus, most untrained or recreationally active males may expect to gain ~ 1.5 kg of FFM when training for 4–24 weeks. When the studies were divided using training status, there were no statistically significant differences in FFM gains between trained and untrained individuals, but the numerical increase was 0.56 kg greater in untrained (untrained: 1.54 kg, 95%CI: 1.12–1.96, vs. the trained: 0.98 kg, 95%CI: 0.17–1.79). On the lower end, the bottom 10% of responders reported

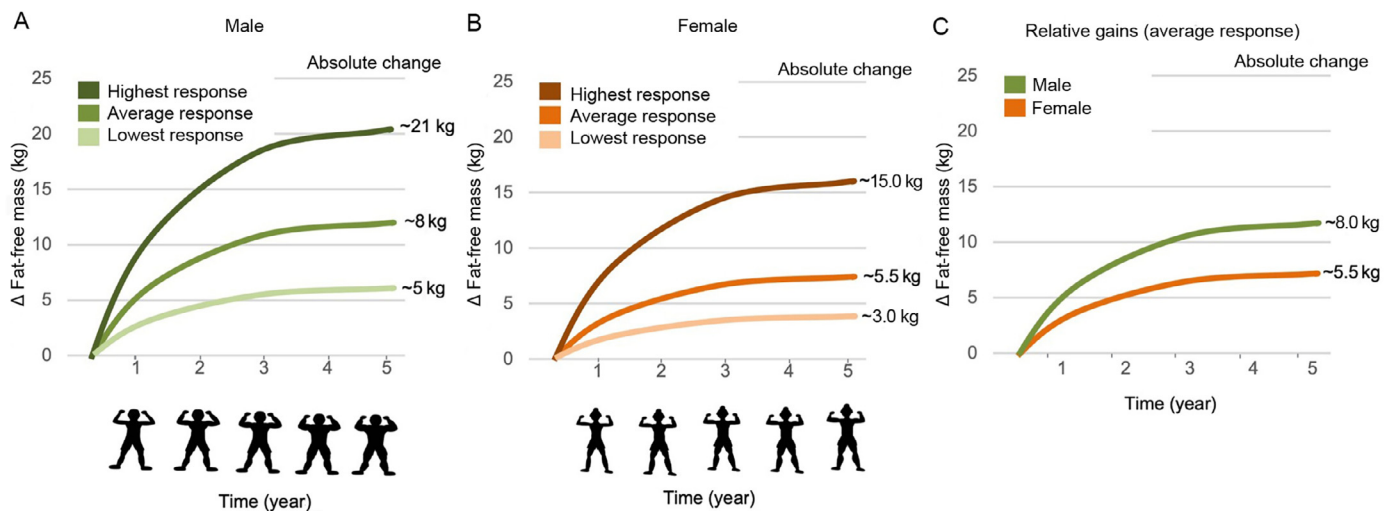


Fig. 4. A schematic representation of the inter-individual variability in fat-free mass gains following long-term resistance training in males and females. (A) Absolute changes in males, illustrating lower, average, and higher responders. (B) Absolute change in females illustrating lower, average, and higher responders. (C) Relative gains (average response) in males and females. This figure is based on concepts from Refs.^{130,134,137} but is original work.

increases of 0.44 kg whereas on the higher end, one study reported an increase of 5.62 kg,¹³⁵ which is significantly greater than the rest of these data, with the highest 10% above 3 kg and the highest 2.5% above 3.8 kg. Thus, it could be considered that higher “responders” for FFM from ~12 weeks of RET would be ~3 kg, a low responder would be ~0.4 kg, and an average responder would be ~1.5 kg. These effects increase substantially with the inclusion of anabolic steroid use, with Bhasin and colleagues¹³⁶ reporting a mean increase of 6.1 kg of FFM following 600 mg of weekly exogenous testosterone after 10 weeks.

When considering the entirety of a lifting career, inferences can be made from examining the range of FFMI in natural bodybuilders, which has been reported between 22.7 and 25.0 kg.¹³⁷ which amounts to 12.5–20.2 kg in total; or ~2.5–4.0 kg on average per year over a 5-year training period.¹³⁷ While gains of 2.5 kg of whole-body FFM per year may seem modest, they reinforce the reality that muscle growth has physiologically constrained limits and understanding these can help set realistic expectations for progress in both novice and experienced lifters. When considering the use of anabolic steroids, Moreno and colleagues¹³⁸ performed a case study in competitive physique athletes, which included an Olympia-qualified pro open bodybuilder. This athlete was 184 cm and weighed 131.4 kg at 12.2% body fat (via bioelectrical impedance analysis), which corresponds to a FFMI value of 34.1 kg.¹³⁸ It is essential to recognize that FFM accounts for all body mass that is not comprised of fat tissue. FFM includes bone, body water, skin, organs, and other tissues, and is not entirely skeletal muscle. However, the primary tissue responsible for driving changes in FFMI over time will be skeletal muscle mass.

Considering that females experience proportional gains relative to males,^{49,50} their absolute gains will be smaller due to lower baseline levels of FFM. Using Benito and

colleagues¹³⁴ for estimation purposes, a baseline male example with ~65 kg of FFM, a 1.5 kg gain over 12 weeks, represents a ~2.3% increase. A long-term increase of 10 kg corresponds to a ~15% increase. Making the comparison to a female with 20 kg less FFM (45 kg of FFM), and applying the same proportional increases, a 1.5 kg gain in males equates to roughly a 1 kg gain in females, while a 10 kg gain in males would be approximately 7 kg in females.¹³⁰ Thus, a reasonable estimate is that absolute FFM gains in females may be ~70% of those observed in males. Fig. 4 provides a schematic representation of the inter-individual responses in fat-free mass gains in both males and females in response to RET.

Understanding the expected rate of progress at any given stage and the potential for long-term growth is an important consideration. This knowledge enables individuals and practitioners to improve their training progress more effectively and gain deeper insights when interpreting RET research. As discussed, we have a limited understanding of inter-individual responses to RET; therefore, some values may overestimate or underestimate an individual’s response. Moreover, these values are not meant to deter or discourage individuals from pursuing their upper genetic ceiling, but rather to encourage realistic expectations and acknowledge the robust response that comes from beginning an RET regimen. We propose that such knowledge may be particularly useful for younger (i.e., <18 years) novice trainees and coaches when designing programs and conveying expectations to athletes and trainees. We do, however, acknowledge that during pubertal growth, our estimates from adults may be subject to a higher upper level. Lastly, while skeletal muscle hypertrophy is not limitless, the benefits of regular RET participation extend far beyond increases in muscle mass.¹³⁹ Such benefits include the preservation of functional independence, reduction in fall risk, improvements in metabolic and cognitive health, and the prevention of age-related disability.¹²⁵

6. Practical application

A recent systematic review and meta-analysis of 3.3 million participants across 32 countries reported that only 1 in 5 adolescents and adults meet the World Health Organization's recommendations for both aerobic and strength activities,¹⁴⁰ highlighting the need to educate the general population that any form of participation in RET will provide increases in strength and hypertrophy when compared to no exercise and this conclusion is indiscriminate of certain RET variables.¹⁴¹ RET programming should emphasize a personalized, goal-dependent approach with relatively high effort, aiming to achieve a proximity to failure (i.e., ~2 reps in reserve), sufficient volume, and progressive overload (i.e., an increase in load, reps, and volume) to maintain a sufficient mechanical tension stimulus. Since hypertrophy can be achieved using a wide range of loading paradigms, individuals can tailor their training to suit their personal preferences, physical limitations and lifestyle. Importantly, RT strategies designed to transiently elevate systemic hormones or intentionally promote metabolic stress offer little additional benefit for hypertrophy. Moreover, while cell swelling and “the pump” may feel satisfying, neither process affects long-term muscular adaptations. Ultimately, RET should be centered on sound programming principles (i.e., adherence, volume, frequency, intensity, *etc.*), as these are by far the most important factors to promote skeletal muscle growth across individuals. Finally, while individual responses to RET exhibit considerable variability, the majority of individuals can experience meaningful improvements in strength and function, with some expectation of increased muscle mass with consistent RET.

7. Conclusion

Skeletal muscle hypertrophy is a complex, multifactorial adaptive process in response to RET, predominantly driven by mechanical tension and the activation of mechanotransductive intracellular signaling cascades (e.g., via mechanotransduction-mediated pathways signaling through mTORC1). While several mechanisms—such as endogenous hormonal elevations, metabolite accumulation, cell swelling and sarcoplasmic hypertrophy—have been proposed as meaningful contributors to hypertrophy, the current evidence does not support these as factors promoting muscle growth.

By critically examining these longstanding myths with mechanistic and applied outcomes, we gain a better understanding of the underpinnings of load-induced skeletal muscle hypertrophy. Evidence-based practice should emphasize the fundamentals of RET: progressive overload, sufficient training volume, a proximity to failure, adequate recovery, and long-term adherence. As one ages, it becomes particularly important to use RET as a powerful tool to mitigate the loss of skeletal muscle and increase healthspan. As individuals progress in their training and the adaptive rates of muscle accretion diminish, it becomes imperative to shift the lens from short-term changes (weeks) to longer-term timelines of months and years. Finally, by providing estimates of short- and long-term muscular adaptations and upper-body hypertrophy,

individuals, coaches, and practitioners can assess rates of progress with greater clarity. As exercise science and RET research evolve, the continued integration of mechanistic insights with applied outcomes will be essential for refining the understanding of skeletal muscle hypertrophy and supporting the development of more effective, individualized training strategies across diverse populations.

Authors' contributions

DWVE conceived of the idea for the paper, drafted the manuscript, and contributed to critical revisions; MJL contributed to the writing and provided critical editing of the manuscript; BW assisted in writing and critical revisions of the manuscript and design of the graphical abstract; JN contributed to writing and provided critical editing of the manuscript; SMP contributed to the conception of the manuscript, contributed to critical revisions, and provided senior oversight. All authors have read and approved the final version of the manuscript, and agree with the order of authorship.

Declaration of competing interest

DWVE is a coach in the fitness industry; JN is the director at STRCNG Incorporated o/a Jeff Nippard Fitness. All the support had no involvement in the study design and writing of the manuscript or the decision to submit it for publication. The other authors declare no conflicts of interest. Given his role as editorial board member, SMP had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

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Supplementary materials

Supplementary materials associated with this article can be found in the online version at [doi:10.1016/j.jshs.2025.101104](https://doi.org/10.1016/j.jshs.2025.101104).

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